

Cosmological Relativity: what the construction delivers, what it declines, and where its edge is

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Abstract

This paper states, in one place, what the Cosmological Relativity corpus establishes, at the weight each result is established at, together with what the construction declines to claim and where its open edge lies. Nothing here is new: every claim is drawn from one of seventeen papers or from the ledgers those papers cite, and is stated at the register its source gives it—a proposition as a proposition, a reading as a reading, a conjecture as a conjecture. What is new is the assembly, and the two things an assembly can supply that a set of papers cannot: the joins, where a result in one paper is load-bearing for a result in another, and the relative size of the frontier against the body of work that surrounds it.

1 What forces the object, what it is, and the distinctions it forces

Cosmological Relativity is a construction on one geometric object, and almost everything below is that object read from somewhere. But the object is not where the argument starts, and presenting it first would ask a reader to grant the thing in question. It is reached: a measurement forces a cosmic foliation, reading that foliation ontologically is the augmentation, and maximal symmetry then leaves exactly one manifold to carry it. This section takes those in order, and closes on the distinction every later result depends on — what a *boundary* of such an object is.

1.1 What the measurement forces

The cosmological redshift is not a property an emitting source carries. It is the path integral of the expansion rate along the photon’s own trajectory, accumulated continuously rather than fixed at emission [12]. That distinction is what makes the isotropy of the microwave background’s monopole a measurement of something other than what it is usually taken to measure.

Grant the common reading its strongest form. Let last scattering be *perfectly* homogeneous. Then the observed temperature contrast is exactly the variation in the integrated expansion, with nothing left over, and *the entire observed isotropy is the statement that the integrated expansion was the same in every direction*. Homogeneity at the source does not *supply* the isotropy; granted perfectly, it removes one term and leaves the isotropy as a direct measurement of uniform expansion [12].

Under the alternative — that the large-scale expansion rate tracks the matter lumpiness region by region, with no single global scale factor — the accumulated redshift would scatter across the sky by $\sim 10^{-3}$, against an observed $\lesssim 3 \times 10^{-6}$: *an exclusion by three orders of magnitude. And every choice in the estimate biases it downward, so the number is a floor rather than an estimate* [12].

Two escapes remain and they are closed separately. The *statistical* escape is closed by the floor. The *structured* escape — a finely tuned, spherically symmetric inhomogeneity centred on the observer, which would give an isotropic redshift with no global uniform expansion and is untouched by any $\sqrt{N}$ argument — is closed by the Copernican principle *together with*

the independently measured isotropy of the expansion history itself. *With both excluded, the observed isotropy forces uniform cosmic expansion.*

What a single datum thereby establishes is worth stating exactly, because it is larger than a parameter bound. The rejected region is not a corner of parameter space but *the entire class of histories lacking a single global scale factor on a single global time.* To reject that class is to select a global, time-ordered expansion — the cosmic foliation, made dynamical. *And the foliation is logically prior to everything the standard model otherwise assumes:* “space” is one of its slices, “expansion” an ordered family of them, isotropy and homogeneity properties of a slice — none of them statable until the foliation is fixed.

1.2 The augmentation that follows, and its structural counterpart

Augmenting general relativity by fixing a physical foliation and reading it ontologically — the lapse the objective rate at which the existing world advances, the shift the relativity of synchrony — is *necessary and sufficient* for a coherent formal description of an existing, evolving world [13]. *Necessary*, because a description lacking the lapse part collapses existence into occurrence. *Sufficient*, because the two parts close the only coherent escapes. No field equation is altered.

The standard reading fills general relativity’s silence about which foliation is physical by collapsing two distinctions at once: it reads the relativity of simultaneity as the absence of any objective present, and it reifies the four-manifold as a thing that exists [12]. §2 takes those apart; the augmentation is what remains when they are held apart.

A second collapse entered with the first cosmology and runs in the opposite direction from what the measurement establishes. Einstein’s static solution fixes the cosmic scale from the content — with $\lambda = \kappa\rho/2$ the radius is $R = 1/\sqrt{\lambda}$, so *the density of the world-matter determines the size of the world* — and Friedmann’s equations carry that direction of determination forward, the scale factor’s evolution sourced by what the universe contains. *What the isotropy forces is the reverse ordering for the rate:* it is fixed independently of how the content is distributed, since a rate tracking the content is precisely what the measurement excludes. *Whether the same reversal extends from the rate to the scale is not settled by the measurement* [12].

And the structure the datum forces was posited, attacked, defended and abandoned within four years. Einstein assumed it in February 1917 in his own words “against the spirit of relativity”, on the empirical ground that stellar proper motions are small compared with c . De Sitter countered a month later, placing the choice among candidate universes outside physical argument altogether and pressing as his sharpest objection that in Einstein’s solution “time has a separate position”. Eddington defended it in 1920 on geometric grounds, in the declared absence of any experimental knowledge on cosmical scales — *he names the absence of the discriminator and selects anyway.* Einstein then went nearly silent on cosmology and never addressed how the cosmic time his own assumption distinguished stands to the relativity of simultaneity [12]. *The assumption was correct, and the objection was correct as a description of the structure though not as a reason to reject it.*

And the same augmentation is forced a second time, from general relativity alone. The event horizon of a collapsing body is a metric singularity met only at infinite exterior time, so no finite layer carries a completed horizon (§1.6) [2]. *These two forcings are independent and co-equal, and neither rests on the other:* one is measured and one is structural, and a reader may take either without the other.

1.3 Three senses of “real”, fixed before the word is used

The corpus uses *real* in three senses and keeps them apart; this paper does the same. They are not degrees of one thing.

Real in the arithmetic sense: a geometry has a real coordinate basis, with no imaginary coordinate doing any work. That is the sense in which the substrate below is a real manifold,

and the claim is about coordinates and signature and nothing more.

Real by construction: a feature is built by a construction and really seen from the vantage that builds it. The perspectival mass is of this kind, and so is the compact face on which a continuous $\mathfrak{su}(3)$ could live.

Existent: the ontological sense, whose criterion in this framework is temporality. *The existent is the evolving three-dimensional layer, and the manifold is its record* [13]. By that criterion the compact face is real by construction but, being Riemannian and atemporal, *not a co-equal existent*—it holds no clock [14].

So nothing in this section claims that a five-dimensional object is what exists. The substrate is the geometric core the construction reads and cuts; what exists, on this framework’s own criterion, is the layer. The higher-dimensional geometry is what the layer is a cut of, not a thing possessing the layer’s kind of being, and the four-dimensional manifold is the record rather than the existent.

1.4 What maximal symmetry then selects

The foliation is forced and the manifold that carries it is not yet fixed, and this is where the construction’s one object arrives — as a selection rather than a premise. The maximally symmetric solutions of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ form a one-parameter family with curvature radius $\alpha = \sqrt{3/|\Lambda|}$. They are usually met as hypersurfaces in a flat five-dimensional space, which is a convenient picture and not the manifold: solving that embedding for the fifth coordinate and substituting back returns an intrinsic four-dimensional line element in which the manifold is coordinatised by four *real* coordinates, real regardless of the sign of the curvature.

The imaginary fifth coordinate is a property of one embedding picture, not of the surface. *This is reality in the first of the three senses above, and in that one only.*

Reading the metric’s eigenvalues off that real line element gives the result the programme’s ontology stands on. Three eigenvalues are positive always, and the fourth’s sign is fixed by real data alone.

Proposition 1 (The unique intrinsically Lorentzian maximally symmetric manifold). *De Sitter space is the only real maximally symmetric manifold carrying an intrinsic Lorentzian signature. It has a real coordinate basis, satisfies the maximal-symmetry requirement, and is the only such manifold with the requisite causal structure* [16, 1].

Two ways the signature can turn are worth separating, because they are of different kinds. At fixed positive curvature the crossing occurs where the fourth eigenvalue passes through a *pole*: its magnitude diverges and its sign flips, the numerator being constant throughout. *So the metric does not degenerate at the crossing*—a signature change through zero would send the determinant to zero and leave no metric there, and that is not what happens. The one place the eigenvalue vanishes is the null cone: not a crossing within a member of the family, but the family’s own singular leaf [16].

That the substrate is reached through imaginary instruments is therefore a fact about the instruments. The embedding coordinate, the equatorial seam’s continuation, and the reassignment at the cosmogenesis are each places the construction uses an imaginary variable and lands on a real manifold; the geometry is intrinsically real at every point they land on [16]. In one case the instrument is not optional and the case is the construction’s own central algebraic object: over the field of the mass parameter the horizon cubic is irreducible, and below the Nariai mass its discriminant is positive, so its three roots are real and distinct. That is exactly the hypothesis pair of *casus irreducibilis*, whose conclusion is that no root lies in any real radical extension—the three real horizon radii admit no expression in real radicals as functions of the mass, and every radical route to them passes through $\mathbb{C}$ [16]. The imaginary is the only available road, and the roots it lands on are real.

1.5 One scale, and what that means

The substrate carries a single dimensionful scale. Read projectively it is a Cayley–Klein geometry^L whose *absolute* is the null cone, and its geodesic separation is the Cayley–Klein log-cross-ratio taken against that absolute, with the Cayley–Klein constant equal to α [16]. So α is not a length inserted into a metric but the constant of a projective geometry whose absolute is the light cone—which is the signature proposition above, read projectively rather than metrically.

The constants through which physics is written on it are unit gauges, each a nameable feature of the substrate: c is the null-ruling slope of the equilateral hyperboloid, Λ the waist, and G enters only as the offset-length GM/c^2 , the mass being the offset of the cut from the central geodesic [16, 10, 4]. On this accounting the gravitational–cosmological–quantum sector spends no free dimensionless constant, and the reason is structural rather than counted: neither real form supplies a second invariant, and a dimensionless magnitude needs two [16].

What the count leaves is not a constant but an initial condition, and the corpus carries it as measured content rather than as a derived quantity—as flat Λ CDM carries the baryon-to-photon ratio from a baryogenesis it does not model [11].

1.6 What a boundary of the substrate is

Here the corpus draws the distinction that everything downstream depends on, and it is a distinction between two kinds of singular locus rather than between a real one and an artefact.

A *metric* singularity is a locus at which a spatial extent contracts to zero while the curvature stays finite. A curvature singularity is a locus at which an invariant of the curvature diverges. The event horizon of a collapsing body is the first kind: a null hypersurface along whose generators the spatial extent also contracts to zero, occurring only as the null future boundary approached at infinite exterior time [2]. Three consequences of that follow immediately, and each is a distinction a reader needs before meeting any application.

Observer-dependence is not the criterion. It is tempting to sort these loci by whether they are seen by every observer, and the sorting fails at once: the Rindler horizon is perspectival and carries a flux, so “perspectival, therefore no flux” is refuted before it starts. What sorts the four horizons the corpus treats is *completion*—whether the boundary is reached on any finite layer—and completion sorts all four [2].

And the finite-curvature case is the one the metric-singularity result treats. It is a different object from the locus standardly called the curvature singularity at $r = 0$, where the invariants computed on the areal radius diverge; the first is established with the event horizon as its exemplar, and nothing is said there about the second beyond noting that it is a different object [2]. *That the two admit a common treatment, and what becomes of the divergence at $r = 0$ under it, is established in the companion* [3, 4]. *No claim of necessity or sufficiency is made here in either direction*: the scope is what is stated, and reading it as a general criterion for when a boundary may be crossed would assert what the source declines to assert.

And formal likeness does not sort the two classes. A chart such as Eddington–Finkelstein parametrises approaches to a singular locus by a coordinate that degenerates there, so a single metric place is drawn spread out as an extended line. Mercator draws the North Pole by the same projective mechanism, and the visual appearance is identical: a place drawn as a line. *But the reason for the appearance is the opposite*. At the pole the chart manufactures an extension that is not there, and a chart centred on the pole collapses the line back to the point it always was. At the horizon the chart spreads out a metric collapse that *is* there: reading the line back yields no ordinary point a better chart would reveal, because no chart removes it—the *collapse is in the metric, not in the projection* [3]. So no argument from how a singular locus is drawn can sort the two, in either direction.

1.7 What follows for the reading, and what does not

Two guards belong here rather than later, because both are misread in the same direction.

Perspectival does not mean unreal. Where the corpus calls a feature perspectival it means built by a construction and really seen from the vantage that builds it—the curvature divergence of a perspectival metric is real as that metric’s, and reading it as an artefact in the pejorative sense is the misreading the epistemology is written to defeat [6, 3].

And identifying a feature as perspectival is not a dismissal of the construction that carries it. Sweeping a radial curve to recover a spatial geometry is the legitimate and indeed the natural way to chart a geometry from a given vantage, and the result agrees with the standard geometry on every observable outside the horizon. What is identified is not an illegitimate operation but a misreading of its product: treating features the construction manufactures—the asymmetric labelling, and with it the curvature singularity at $r = 0$ —as features of the geometry rather than of the chart. *The construction is faithful as a description from its vantage; the error is ontological* [3].

2 What the forcings pry apart

The forcings of §1 do their work by separating pairs of ideas the standard reading holds fused. *Each separation is a result rather than a clarification:* in every case the fused form is shown to make a claim the evidence refuses or the formalism cannot carry, and the separated form is what the rest of the construction is written on. This section states the pairs, because a reader who keeps any of them fused will read §§3–7 as claiming something other than what they claim.

2.1 Literal and perspectival

The appearances divide into components a projection images faithfully and components that are artefacts of the projection, and *both classes are non-empty* [6]. The Sun’s annual path along the ecliptic is a perspectival illusion of the Earth’s motion; the Moon’s monthly circuit is a literal orbit. Because both occur, *no blanket reading is admissible* — and reading every appearance literally is the first thing a method must forbid.

A perspectival component need not be an illusion, and the corpus’s own case is the sharp one. The metric’s verdict that two null-separated horizon events share one place and one instant is a genuine fact about the measure — the separation truly collapses — *yet perspectival still*, a fact about the ruler and not about the identity of the events, which stay distinct on the point-set the ruler is laid over [2]. *So the perspectival class is not the class of false appearances but of those that image the projection rather than the world*, and some are entirely real as facts about the projection.

Nor does formal likeness sort the two, which is why the separation cannot be done by inspection: a genuine metric collapse drawn as an extended line and the Mercator map’s rendering of the pole as its top edge are identical in form, and only exhibiting the projection each arises under sorts one from the other [3].

2.2 Saving an appearance and explaining it

An admissible world must *exhibit* the projection under which a perspectival appearance arises and show why it appears as it does — *not discard it, and not merely reproduce it* [6]. A model can reproduce the ecliptic path exactly while saying nothing true about why it appears. *The retrograde loops are not real reversals to be tracked by finer machinery but the parallax of the Earth’s own motion*, and the right theory dissolves the puzzle by identifying what the appearance is.

And a dissolution can be entirely ontology-free, which is worth having before §5. The continuation through the curvature singularity undoes the standard verdict not by any claim about which manifold is fundamental but by exhibiting the continuation that carries the curve through — *the puzzle dissolved by identity, nothing assumed about the world beyond the appearance’s own smoothness* [3].

2.3 Existence and occurrence

Events occur; the world exists. To deny the distinction — to hold the four-manifold itself the existent — is to treat occurrences as existents, which on analysis smuggles a fifth, meta-temporal dimension along which the block is supposed to “exist”, and is incoherent as a basic description [12]. *The regress is the argument*: whatever endures does so through some time, so a four-manifold said to exist requires a further time it endures through.

This is the distinction §1’s third sense of “real” turns on, and the lapse is its formal counterpart: *the rate of advance of the layer that exists, which is what a clock measures*.

2.4 Simultaneity and synchrony

The relativity of synchrony is general — a feature of relative motion, the light postulate fixing only its magnitude — and it is the *shift*, the freedom of a moving observer to tilt the slices off the comoving normal. *Simultaneity is the separate question*.

To read the relativity of synchrony as the denial of an objective simultaneity is a modal fallacy [12, 6]. From the premise that the appearances contain no local discriminator between two candidate worlds, it does not follow that the worlds are identical, nor that the structure distinguishing them does not exist: *the absence of a local test is not the absence of the fact*.

The phrase is exact rather than figurative. *Locally flat* is the technical condition that every point of an embedding has a neighbourhood in which it is standard, and high-dimensional topology turns on locally flat against wild embeddings precisely because local standardness leaves the global invariant free: a submanifold can be locally indistinguishable from the trivial one at every point and globally knotted. *So the modal fallacy is the inference from local triviality to global triviality*, and that inference has a standing counterexample in every dimension where the question has been asked [6].

The history is a record of the fallacy committed and corrected. No stellar parallax was detectable for two millennia and this was read as telling against the Earth’s motion — though before the cosmos was given depth the relevant parallax was not a conceivable measurement at all. No local experiment distinguishes rest from uniform motion, and this was read as the non-existence of any preferred rest. *In each case the missing discriminator was mistaken for a missing fact, and a later non-local measurement supplied both* [6]. *An absent discriminator is not an absent fact*, and whether a discriminator is even available is a question about the picture in hand rather than about the world.

2.5 The map and the territory: a direction of construction

The separations above fix a direction, and running it backward is the standing error of which reading every appearance literally is the local symptom. *The valid order builds the ontology from the evidence, the kinematics from the ontology, and the coordinates from the kinematics* [6]. The reification error runs the chain backward: it takes the coordinate system as given and reads an ontology off it, treating the map as the territory and an apparent symmetry of the description as a fact about the world.

Applied to a mature mathematical theory the demand is sharper than it first appears, because there the appearances include the formal structures one has derived. *A four-dimensional manifold that organises the record of events is a coordinate scaffold of this kind*; granting it

existence runs the chain backward, and the discipline forbids it on the same ground it forbids reading the ecliptic as a real solar orbit.

And the reification can hide in something smaller than a manifold — in a verb tense. The founding literature calls a black hole a “collapsed” object, the past participle quietly reading horizon formation into the causal past of an exterior observer, where the causal structure places it only at the infinite-future boundary. *The grammar must follow the geometry* [2].

2.6 Requiring a phenomenon and permitting it

The last pair is a criterion rather than a distinction between things, and it is the one the rest of this paper is weighed on. *Prefer the world that requires the observed phenomena as a consequence of its structure to the world that merely permits them through adjustable parameters* [6]. A framework on which the phenomenon could not have been otherwise is an explanation; one on which it is a tuned possibility among many is a description.

In the ontological register this has a sharp form. A structure carrying an unforced parameter — a modulus fixing how some symmetry is broken, which neither the appearances nor any deeper principle pins — *is not a single world but a family*, and answers “what is the world?” with a family rather than a world. *It is inadmissible as it stands rather than merely disfavoured.* The maximally symmetric structure is the unique one carrying no such modulus: every less symmetric structure requires a choice of how to break the symmetry, and maximal symmetry leaves nothing to choose [6].

That clause is a statement about a group action and it has an exact dynamical counterpart. What leaves nothing to choose is a group acting *transitively*: a modulus is a coordinate transverse to the orbits, so one exists precisely when the action is not transitive. *Read on the substrate’s geodesics the same transitivity says the flow is maximally superintegrable* — the isometry group acts transitively on the unit tangent bundle, so every geodesic is every other seen from another vantage [16]. *Least-arbitrariness and superintegrability are one property read epistemically and dynamically*, and a modulus appearing is the same event as the action ceasing to be transitive.

This is why §1’s selection of the substrate is a selection and not a preference: the alternatives are not worse worlds but families of worlds, and the criterion that excludes them is the same one by which any theory predicting a datum is preferred to one accommodating it.

3 What is derived from the object

With the substrate fixed, the construction’s working object is a *cut* of it, and one instrument generates the cuts. This section states what that instrument returns, what it cannot reach, and where its boundary falls — and it separates, at each step, the theorems from the reading they ground.

3.1 Straight cuts are vacuum

In the construction’s gauge the vacuum condition $T_{\mu\nu} = 0$ is not a system to be solved but a single first-order linear ordinary differential equation, $rf' + f - 1 + \Lambda r^2 = 0$, whose *entire* solution space is

$$f = 1 - \frac{2M}{r} - \frac{\Lambda r^2}{3}, \quad (1)$$

the whole Schwarzschild–de Sitter family, with M the single constant of integration [10]. *The vacuum sector is exactly the kernel of the matter functional, derived rather than matched:* straight cuts are vacuum. Taken with the bend identity below — which shows every density is realised by some cut, so the functional is onto — the two give a kernel of dimension one over a vanishing cokernel, an index of 1, generated by that constant.

Two things follow that are worth having early. There is no room in the vacuum family for a third term: the solution space is the whole of it, so a rate written on this kernel carries Λ and the cut's offset and nothing else. And the question this answers is not a new one — it is the question the programme's own 2012 dissertation left open, asking what, if not the world-matter, fixes the geometry; the kernel is the answer, and it is one the corpus states in the asking author's terms.

3.2 Matter is the bend of the cut

Writing any curve as a departure from the vacuum profile, $f = 1 - 2m(r)/r - \Lambda r^2/3$, gives $8\pi T^t_t = -2m'(r)/r^2$: the energy density $\rho = m'(r)/4\pi r^2$ is the radial growth-rate of the enclosed mass, the bend of the curve off the constant- M profile [10]. Checked on a non-vacuum member, Reissner–Nordström–de Sitter is the bend $2m(r) = 2M - q^2/r$, returning the electromagnetic stress-energy off its q^2/r^2 term.

And the reading is not an artefact of the symmetry, which is the objection it would otherwise invite. For a general leaf the density is the leaf's intrinsic-curvature departure from the substrate, $16\pi\rho = {}^3R + K^2 - K_{ij}K^{ij} - 2\Lambda$ — which is the Hamiltonian constraint, the spherical identity being its symmetric case [10].

What the identity states is what the bend is, not why a leaf bends as it does. That is the matter content's own generative law — a dynamics for the curve itself, as against the ordinary leaf evolution that carries it — and the construction does not supply one. *The source names it as the deepest question the construction opens onto* [10], and it is carried on the frontier below.

3.3 What the instrument reaches, and the wall

The reach is a theorem: *the range of the operator is the symmetry-reducible sector of general relativity* [8]. The size of what is reachable is set by how much symmetry a class spends — a finite parameter family where the class reduces to ordinary differential equations, a functional family where it remains a partial differential equation problem — and the reachable sector is the constrained, non-radiative skeleton of the theory.

The boundary of that reach has a positive identity, and stating it the obvious way gets it wrong. The wall is the loss of isometry, and what lies past it is *the onset of free gravitational radiation*: the graviton's two propagating polarisations, read on the matter side as dynamical inhomogeneous sources [8]. *So the wall is a seam and not a defect* — it is where generation by symmetry hands off to ordinary inhomogeneous evolution, and the handoff is a feature of the classification rather than a failure of the instrument.

One thing this section does not rest on is worth naming, because it is the part that looks like the result and is not. Within a reachable class the operator's four data — leaf, lapse, shift, vantage — supply exactly the functions the general invariant metric admits, so the cut spans the class. *P9 says plainly that this surjectivity is not the content*: once the cut carries the class's function count, the cut ansatz is the general invariant metric and spanning is near-tautological. *The content is the identification of the vacuum members as the substrate's own family — the kernel — and of matter as the bend.*

3.4 General relativity's constraint algebra is the substrate's grading

At the symmetric cut the substrate's algebra splits as $\mathfrak{so}(5, 1) = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(4, 1)$, and all three symmetric-space inclusions hold, $\mathfrak{m}$ not being a subalgebra. Under the correspondence between $\mathfrak{m}$ and the normal deformations and between $\mathfrak{h}$ and the tangential ones, *these are the hypersurface-deformation brackets term for term* [9].

So the algebraic shape that makes the Dirac algebra puzzling — two normal deformations bracketing into a tangential one — is exactly the shape of a symmetric-space coset: two coset

directions bracketing into the isotropy. They are the same grading. *The claim is a recognition rather than an addition*: no structure is added to general relativity, and what is exhibited is that a pattern the theory already carries is the substrate's.

The coefficient that makes the algebra an algebroid rather than an algebra — the structure *function* standardly identified as the canonical root of the problem of time — is, on the symmetry-reducible reduction, the coset metric of that symmetric space [9]. *And the paper states at once that this is not a naive tensor equality*: the structure function is the Riemannian inverse spatial three-metric, the coset metric the Lorentzian five-dimensional form of signature $(1,4)$, and what is identified is the *reduced* structure function on the symmetric-cut pattern with that coset form.

3.5 What is a theorem here and what is the reading

The construction's covariance claim is that general relativity holds *a single geometry* invariant under change of chart, while this construction holds *the substrate* invariant under change of *geometry*, with the slicing curve as the gauge object: *“many line elements, one geometry” becomes “many geometries, one substrate.”*

That is a reading, and the source labels it as one. The theorems — the vacuum kernel, the bend-density identity, the lapse split, the cosmological geodesic, the second-ruling embedding — are computed and verified; *the covariance reading is adopted on their strength and explicitly does not follow from them as a corollary follows from a theorem* [10]. It is stated here at that weight and at no other.

4 What is computed, and what it is confronted with

The construction meets data in three places: a rate, a set of light-element abundances, and the microwave background's angular structure. Each is stated here with the register its source gives it, and the third includes a measured disagreement for which no mechanism is in hand.

4.1 The rate, and what its two parameters are

The rate's two parameters are the substrate's curvature radius and the *offset of the cut*, and *neither is a content of the universe* [11]. The offset is measured directly and calibration-free — without a distance ladder, without the microwave background, and without any density.

The familiar cosmological form is then a coincidence of form, and the distinction matters more than it first appears. Written in the fitted pair, the same rate reads as the Friedmann one, with the density parameters following identically. *That is a translation between parameter sets and not a decomposition into components*: the two readings agree on the function and share nothing beneath it. A reader who takes the agreement as a decomposition has imported the very thing the construction declines to assume.

4.2 Three levels, and why a mis-assignment is fatal

Every rate-bearing quantity in the construction sits on one of three levels, and the assignment is not a convention [11]. *The foliation stacking rate* is read leftward — the cut is primary and a density is the name of its bend rather than its cause — and it is empirically forced rather than modelled. *The leaf-level local dynamics* is the ordinary Friedmann readout with radiation gravitating normally: the same single geodesic, read *inward* as dust collapse where the first reads it *outward* as cosmology. *The projection* is what an observer reads.

There is no boundary in time between the first two levels. The distinction is kinematic and holds at every epoch, the decomposition exact, the two rates differing by the radiation term

alone. *A quantity assigned to the wrong level is not approximately right*; it is a different physical claim, and §4.5 records a case where the corpus itself made that error and what it cost.

4.3 The abundances, and the miss that discriminates

The cooling leg *is* a standard big-bang nucleosynthesis — cooling through the window at the standard rate from a fully dissociated start — and a genuine multi-nuclide network integrated explicitly on that history returns $Y_p = 0.247$ against an observed 0.245, $D/H = 2.51 \times 10^{-5}$ against 2.53×10^{-5} , ${}^3\text{He}$ at 1.05×10^{-5} , and ${}^7\text{Li}$ at the standard several-fold over-prediction — *all jointly from the single inherited η* , at the baryon density the microwave background reads from the peak heights, with deuterium at -0.5σ and helium-4 at $+0.5\sigma$ [15].

And the lithium miss is the discriminating result rather than an embarrassment to be noted and passed over. It is the standard over-prediction, reproduced — which is to say the construction inherits the standard problem exactly, neither curing it nor worsening it. *A mechanism that changed the early history would have moved lithium*; this one does not, because the window and the rate through it are the standard ones.

4.4 The low-multipole floor, and why it does not discriminate

Because the distance slicing is exactly flat while the cosmological layers are a closed three-sphere, the photons are projected through the *flat* geometry while only the *source* modes carry the closed quantisation — *not the hyperspherical transfer of a literal closed universe*, which would deliver the lowest mode to the quadrupole and no deficit at all. The result is a *parameter-free* deficit below $\ell \approx 8$, bottoming at $\ell = 4$ and *not* at the quadrupole [11].

The register here is split and the split is the honest part: the existence, location and minimum are established; *the depth is open*, two transfers differing by up to a factor of two. And the feature does not discriminate between frameworks — it is a prediction the construction makes for free, not a test it passes and a rival fails.

4.5 The acoustic comparison, and a deficit with no mechanism

The acoustic *scale* is met, and the corpus records what meeting it cost: it is an accommodation, and it is spent. *The peak structure is a separate confrontation and it does not come out even.*

Run at converged wavenumber on the leaf congruence — the rate the level rule assigns the perturbations to — with the instrument calibrated in the same configuration against a standard code, the construction returns a first peak at $\ell_1 = 204$ and $\ell_1/\ell_A = 0.6764$ against the sky's 0.7312: *a 7.5% deficit in position*, with height ratios $P_1/P_2 = 1.759$ and $P_1/P_3 = 1.612$ against 2.217 and 2.277. *Two wavenumber cutoffs agree to every digit, and the deficit is neither of the two instrument faults* that a reader would reasonably suspect — the truncation and the projection path are both removed in that configuration.

The seam datum's own freedoms are measured rather than left as an escape. Across the seventeen readings admitted by a four-peak criterion fixed before the numbers, the first peak spans 148 to 228 — a factor of 1.541, and 1.118 across the numeric-phase readings alone — so *the position is a statement of the construction and not a free choice*, and the sky's value lies inside that span. The same holds separately for the spacing, the phase intercept and the alternation ratio. *But a span containing the sky on each statistic one at a time is much weaker than a reading that reproduces the sky, and there is no such reading.*

What the readings do carry is a correlation, and it is the sharpest thing the scan returns. Position and alternation move against each other across the datum, at Spearman $\rho = +0.782$ and $p = 2.1 \times 10^{-4}$: as the datum carries the peak up toward the observed value, the second gap turns from contracting to expanding. Of the readings landing the first peak within one grid step of the sky's, none contracts; of those that contract, every one sits at $\ell_1 \leq 212$. *So the seam*

datum can buy the position or the alternation and not both, and no mechanism for that trade is in hand.

This is where the level rule earns its statement, because the deficit's most natural explanation turns on it. It is tempting to say that the acoustic modes re-enter above the plasma's onset, so that none of them is driven and the comb should be uniform for that reason. *That census is taken on the stacking rate, and the rule assigns the perturbations to the leaf,* which carries a radiation term: on the leaf the band containing the first peak enters the horizon *while radiation dominates*. Those modes are driven, and the driving's size is measured — removing it moves the first peak by 136 multipoles.

So the deficit stands as a measurement without a mechanism. The construction's own level assignment rules out the explanation that would have covered it, and what remains is a number, its size, and the two places it is not: not the wavenumber truncation, and not the projection path.

5 What is dissolved

A dissolution is not a solution. Where a solution supplies a mechanism that answers a question, a dissolution shows the question rested on a premise the construction does not carry — so it is discharged rather than answered, and nothing is owed in its place. The cluster below is presented *graded*, because the tiers differ in what an objector must dispute, and a flat list would hand a reader the whole set to reject at once [13].

5.1 Tier one: what carries the layered reading's full weight

These stand or fall with the reading of §1, and an objector who declines that reading declines these with it.

The horizon-singularity family. The event horizon of a collapsing body is a metric singularity met only at infinite exterior time, so no completed horizon is realised on any finite layer. Three results follow, and each is discharged for a different reason.

Penrose's theorem is a correct mathematical result whose hypothesis is a *realised* closed trapped surface interior to the event horizon. With no completed horizon there is no trapped surface it would enclose. *Singularities are not avoided by new physics*; they are rendered physically irrelevant by causal structure alone, the curvature singularity remaining a feature of a global extension the realised worldtube never instantiates — and cosmic censorship is correspondingly unneeded rather than proved [2].

The Bogoliubov construction behind horizon radiation requires a globally defined horizon, a completed causal structure joining past to future null infinity across it, and permanent loss of causal contact rendering the two vacua inequivalent. *The third has an exact criterion*: two Fock representations are unitarily equivalent precisely when β is Hilbert-Schmidt, and a thermal β fails that test at the *infrared* end, its $1/\omega$ tail making the norm logarithmically divergent [2]. *This is not the claim that black holes do not radiate*; it is the claim that the construction's third premise is unmet on a realised worldtube.

The information paradox arises only if a completed horizon forms and subsequently evaporates. *Both premises fail together*: no completed horizon forms, and with no horizon-induced radiation there is no evaporation to carry the loss. The realised spacetime remains globally connected, with a global Cauchy surface, no hidden interior sector to trace over, and unitary evolution unobstructed. *The paradox is not resolved by a mechanism recovering information — it does not arise* [2].

And the criterion that sorts these is completion, not perspective. An argument of the form “the horizon is perspectival, therefore no thermal flux” is refuted before it starts and is not the argument made: the Rindler horizon is observer-dependent *and complete* — the boost field an

exact Killing field of the whole spacetime — and the Unruh spectrum is thermal. Across the four horizons the programme names, one criterion sorts all four, and it is completion [2].

The problem of time, the hole argument, and closed timelike curves sit in this tier for the same reason: each turns on whether a foliation is a convention, and the augmentation of §1 makes it a physical structure. Where the foliation is physical, the canonical frozen-time problem is a question about a convention rather than about the world; the hole argument’s two solutions are the same world read twice; and a closed timelike curve is not a trajectory an existing layer can carry.

5.2 Tier two: what follows from the single scale alone

These require less. *An objector may reject the layered reading entirely and still owe an answer to them*, because they follow from the substrate carrying one scale and nothing further.

The fine-tuning pair — the cosmological constant problem and the coincidence problem — is discharged by the accounting of §1: Λ is the substrate’s waist rather than a vacuum energy to be computed and matched, and the constants through which physics is written on it are unit gauges. *The question “why is the vacuum energy so small” presupposes that the quantity is an energy*, which is the premise the one-scale reading does not carry.

The local–cosmic boundary is the second: the locus where the local bend exactly cancels the substrate’s cosmological term is derived rather than posited, and it stands in a fixed ratio to the expansion’s own amplitude for every mass [16]. *So the threshold separating what expands from what stays bound is a length the construction computes*, not a density it fits.

5.3 The cosmological puzzles

These are a separate family with a separate home, and they are not graded by the tiers above — they follow from the rate of §4 rather than from the layered reading.

The reckoning is drawn plainly at source: the standard account assembles dark energy, inflation for causal contact and flatness and coherence and scale invariance, and further physics for the tensions; *the construction reads each off Λ* [11]. The horizon and flatness problems are not solved by an early mechanism but do not arise on a rate whose uniformity is forced rather than assumed. The big-bang singularity is a coordinate feature of the reading rather than a breakdown of the theory, the minimum radius being finite. The coincidence problem is dissolved by the same accounting that discharges the fine-tuning pair.

And the oldest of these is the one the construction answers most directly, so it is stated first rather than folded into the list. In any model whose expansion is driven by its contents, matter and radiation decelerate it, so such a universe expands at late times only because it was already expanding, and the expansion is an inertial legacy of an initial condition at which the description breaks down. *Eddington put the objection plainly and Hoyle shared it*: postulating that the velocities were there from the beginning can scarcely be called an explanation of them.

On this construction the acceleration is not a sector at all. On the energy family $(dr/d\tilde{\tau})^2 = E^2 - f$, so $d^2r/d\tilde{\tau}^2 = -f'/2 = rK_G$: *the comoving acceleration is the existent slice’s own Gaussian curvature*, up to the positive factor r . *So the expansion decelerates while the slice is negatively curved, turns over exactly where it is flat, and accelerates thereafter — with nothing added to make it do so* [11].

And the locus of that turnover is the same radius, for every mass, at which a bound structure’s hold gives way to the flow — which is a convergence large enough to state on its own, and §6 states it.

The coincidence problem goes with the same accounting. The present matter density and the dark-energy fraction are *readings of one cosmic clock rather than drivers of the expansion*, so the coincidence of the two dissolves into our observing at a time of order the geometry’s one timescale [11].

And the Hubble tension is placed in a different category from the rest, which is the honest part. It is carried not as a dissolution but as a discriminating datum — the standing here being theory-choice-favoured rather than settled [11]. A reader should not count it with the others.

5.4 The unification divides, and the matter sector

Two further families are carried at lower altitude than anything above, and the corpus says so.

The gravity–gauge and general-relativity–quantum divides are addressed where the boundary paper draws its synthesis: the pieces are established and *the world-correspondence is held open*, the family weighed by the coherence of what it assembles rather than asserted [14].

The matter-sector family — generation replication, the origin of colour, the family-by-chirality structure, the mass hierarchy, matter and antimatter, CPT — is addressed in the matter sector’s own scope section: generation replication read as the three hinges, the family symmetry as the substrate’s own Weyl group, gauged chirality against global flavour as the on-versus-tangent distinction, and matter against antimatter as the conjugation [7]. *These are held at forced-within-CR coherence with the content external, and the corpus records this as the lowest altitude it carries.* §7 states what the same sector declines, and the two should be read together.

5.5 The closure: one substrate read many ways

The meta-collector, where the rest close. The substrate’s maximal symmetry is worn seven ways, and the physics is read as broken-symmetry shadows of one object [16]. *Its status is stated in §1’s terms:* the seven results are each established, and *their unification is the thesis rather than a further theorem* — a conjecture decidable by a named test, and carried here at that weight.

5.6 Tier three: what stands on the bare analytic structure

This tier is untouched by any verdict about what exists, which is what makes it the most robust and the least often noticed. It requires neither the layered reading nor the one-scale accounting — only the analytic structure of the solutions.

The continuation through the curvature singularity is of this kind, as is the identical analytic type of the two critical points. *And the Kretschmann divergence is a pole of finite order* — twelfth in the cycloid parameter, raised from sixth by the chain rule — *which is the whole of why the continuation is available:* a pole is continuable where an essential singularity would not be [3]. The overcritical regime is reached by the same continuation that joins the seam.

An objector who grants nothing else still faces this tier, and that is the reason the cluster is graded rather than listed. Rejecting the reading of §1 costs tier one and leaves tiers two and three standing; rejecting the one-scale accounting costs tier two and leaves tier three. *There is no single objection that takes all three.*

6 One radius, read seven ways

This is the corpus’s clearest convergence, and it is worth setting out whole because no single paper contains it. A boundary observed around every bound structure, and the epoch at which the universe stops decelerating, are on this construction *the same fact at different masses* — and the geometry that makes them one is visible on the substrate itself.

6.1 The locus

For the marginally-bound congruence that *is* this cosmology, $(dr/d\tilde{\tau})^2 = E^2 - f$, so

$$\frac{d^2r}{d\tilde{\tau}^2} = -\frac{f'}{2} = r K_G, \quad (2)$$

with $K_G = 1/\alpha^2 - M/r^3$ the slicing surface’s Gaussian curvature: *the comoving acceleration is the slice’s own intrinsic curvature*, up to the positive factor r [4, 13]. Both sides vanish at

$$r^3 = M\alpha^2, \quad (3)$$

generally in the mass. That radius is the Hubble–Eddington radius, $r_{\text{HE}} = (3GM/\Lambda c^2)^{1/3}$.

So one equation says two things that are not obviously about each other. The expansion decelerates while the slice is negatively curved, turns over exactly where the slice is flat, and accelerates thereafter. And that same flat locus is *the boundary within which local structure stays bound against the cosmic flow and beyond which it is carried off* — the radius the largest bound structures are observed not to exceed [11].

6.2 Why this is a convergence and not a restatement

Two independent quantities change sign at that radius, and the construction does not relate them. The slice’s intrinsic curvature is geometric; the acceleration of the areal radius is dynamical; *nothing connects them beyond their both being read off the same curve* [6]. A geometric identity restated in dynamical words would be one fact twice. This is not that.

And the corpus reads the radius seven distinct ways, in papers written for unrelated ends: the Hubble–Eddington radius proper; the flat locus of the slice’s own curvature; the handover from the sub-marginal bound orbits to the marginally-bound congruence that is this cosmology; the local range of the substrate’s single scale, the one Λ that also sets the global expansion; the acceleration turn; the unique fixed point of the vantage involution; and a consequence of the layered closure itself [6, 13]. *No single paper lists them all* — five are gathered in one place and five in another, three of them shared — *and the convergence is visible only from outside any one of them.*

6.3 The picture on the substrate

Drawn on the hyperboloid the two behaviours are one curve. The substrate is finite at its equatorial throat and opens toward the poles. A slicing line at offset r_0 meets the throat circle rather than passing through it — the manifold is not there at the centre — so the only continuation available is around the circle, and the chart’s $r = 0$ sits on the throat at the back, diametrically opposite the hinge [4]. *The lap runs around the throat and then out along the horn.*

The rate along it separates the loci exactly. Since $(dr/d\tilde{\tau})^2 = 2M/r + r^2/\alpha^2$, the rate vanishes at the comoving turnaround, diverges as $r \rightarrow 0$, and *passes a minimum at* $r = (M\alpha^2)^{1/3}$ — the locus of (3) [13]. *So the re-expansion is slowest exactly there*, and what (2) adds is that the same point is where deceleration becomes acceleration.

On the forced member the locus coincides with the front seam, where the horizon cubic’s double root makes f and f' vanish together; there the bead attains $dr/ds = 1$ with $d^2r/ds^2 = 0$, its unique inflection. *On other masses the two separate with the roots* [13, 4] — *which is why the radius is a statement about every structure and the seam is a statement about one.*

6.4 What this dissolves, and what the standard account has instead

The observed phenomenon has a name and a long-standing awkwardness. A galaxy at the edge of its cluster is said to “enter the Hubble flow” and be carried off, and the cluster to evaporate

at its boundary. *In an account where the expansion is driven by the contents, there is no mechanism for that edge* — only a balance between an attractive term and a repulsive one, meeting at a radius where their magnitudes happen to agree. *A coincidence of forces is a place, not a reason*, and the picture of a structure shedding its outskirts across a threshold has had no better ground than that for a century.

The radius itself is not new and the corpus does not claim it. It is derived in the standard framework as a maximum turnaround radius and proposed there as an independent local test of Λ [5], with the same $M^{1/3}$ scaling. *What differs is what the radius is.* On this construction it is not where two terms balance but *where one quantity changes sign* — the slice’s own intrinsic curvature, whose sign is the boundedness–expansion dichotomy. *So the threshold is sharp because a sign is sharp*, and the structure does not shed its outskirts against a competing pull; the curve those outskirts are carried along simply turns.

6.5 What the appearance is a shadow of

Nothing accelerates anything at that radius. The sign of $d^2r/d\tilde{\tau}^2$ is a property of a fixed curve, read where f' passes through zero, and the deceleration and the acceleration are what continuous motion along that curve looks like from within [6].

So the boundary’s appearance — a coincidence of forces, a structure’s hold giving way to a flow — is shown what it is a shadow of, and not merely declared to be one: a curve whose second derivative changes sign, seen by an observer carried along it. *That is the reading of §2 carried out on an observed phenomenon*, and the phenomenon it explains is the local–cosmic boundary itself.

One further reading marks what the coincidence is not. On the forced member that radius is the front seam, which the stratification marks as the one locus crossed at unit speed *without* dividing two causal regions — f touching zero without changing sign. *So the areal acceleration changes sign there and the causal character does not:* two independent sign-questions answered oppositely at one radius, and an observer crossing it records a change in their expansion history and none in the character of their own radial coordinate [6].

7 What it declines to claim

The declines below are not gaps awaiting work. Each is a place where the construction has an argument for why the thing it does not deliver *cannot* be delivered on the structure it has — and stating them with their registers is what keeps the deliveries of §§3–5 legible as deliveries.

7.1 Colour is not a geometric isometry, and the obstruction is complete

The substrate’s continuous symmetry cannot carry the strong force, and the reason is exact: $\mathfrak{su}(3)$ is not a subalgebra of $\mathfrak{so}(5,1)$, *precisely because the isometry is exhausted* [14]. *Every candidate substrate bundle falls with it:* the algebra needs a complex rank-three module, and a real bundle’s complexification carries a parallel conjugation whose holonomy lands back in the real form.

So the question was never which real bundle, but where the complex structure is — and it is at the branch point. Taking the wall’s definition literally there are three branch loci, one per vantage, and the three wall monodromies with the hinge three-cycle generate $SU(3)$ itself, a full lap being the centre [7].

What that delivers is colour’s discrete content, and it is delivered. What it does not deliver is a force. *The flatness is a complete obstruction rather than a stage not yet reached:* a flat bundle has no curvature to be a field strength, and the corpus shows this rather than conceding it [7]. *A reader should not expect a later section to supply the gauge dynamics; there is an argument here that it cannot be supplied on this structure.*

And the antisymmetry that makes the baryon work comes from the same discipline: it is supplied *not by a gauge datum* but by the three modes being the kernel of one Dirac operator — hence identical particles, so a three-fermion state lives in Λ^3 and *the surplus invariants do not exist*. *The diquark is then a prediction rather than a fit*, nothing having asked that the two-fold carry no invariant.

7.2 The count is computed; the hierarchy is not

The generation number is three, as a γ^5 -graded index of wall-localised zero modes [7]. *The count is computed rather than traced*, and it is the sector’s strongest single result.

The mass hierarchy is not delivered, and the corpus states the boundary rather than blurring it. A cubic supplies the *count* and not the spacing: three roots are three, and nothing in the count fixes the ratios between them. *So a reader who takes the generation result as a step toward the mass spectrum has crossed a line the source draws explicitly*.

7.3 What is held open, and at what altitude

The world-correspondence is held open, and this is the decline that matters most for how the rest is read [14]. The matter-sector family of §5 is carried at *forced-within-CR* coherence with the content external — the corpus’s own description, and the lowest altitude it assigns to anything.

And the closure of §5 is a conjecture and is labelled one. The seven ways the substrate wears its symmetry are each established; *their unification is the thesis*, decidable by a named test and not established by the seven results individually [16]. *Asserting the unification on the strength of its parts would be the one over-claim this corpus is most exposed to*, and the grading of §5 exists partly to prevent it.

7.4 Why the declines are discipline rather than modesty

Each decline above buys something. The colour obstruction is what makes the discrete residue the single geometric opening — had the continuous symmetry been available, the residue would be one route among several rather than the only one. The hierarchy boundary is what lets the generation count be read as an index rather than as a fit. And the world-correspondence being held open is what allows the coherence claims to be stated at all without asserting more than the construction shows.

These are not places the corpus stopped; they are places it drew a line and then used the line.

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Appendix L The Knowledge Ledgers

Each claim marked ^L in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

QUADRIC_GEOMETRY_LEDGER.md

[field bake]

projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes